English and Telugu FastSpeech2-Style TTS Prototype

Technical Report | CPU-Based Proof-of-Concept Implementation

1. Project Objective

The objective is to develop an English and Telugu text-to-speech prototype using a FastSpeech2-style architecture.

2. Dataset

Language	Source	Train	Validation
English	ai4bharat/Svarah	90	10
Telugu	SPRINGLab/IndicTTS_Telugu	90	10

3. Preprocessing

Audio was resampled to 22050 Hz. 80-bin log-mel spectrograms were extracted using 1024-point FFT and 256-sample hop length. Phonemes were obtained using espeak-ng. Durations were estimated by approximately uniform allocation rather than forced alignment. Pitch and energy features were extracted and aggregated to phoneme level.

4. Model Architecture

This simplified model consists of phoneme embeddings, a Transformer encoder, duration/pitch/energy predictors, pitch and energy embeddings, a duration-based length regulator, a Transformer decoder and an 80-bin mel-spectrogram projection. The hidden dimension is 96, with one encoder layer and one decoder layer.

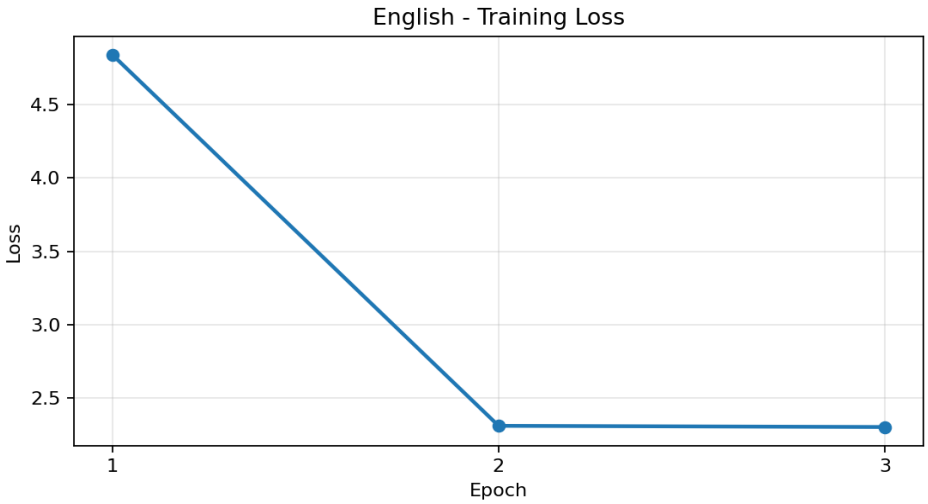
5. Training Configuration

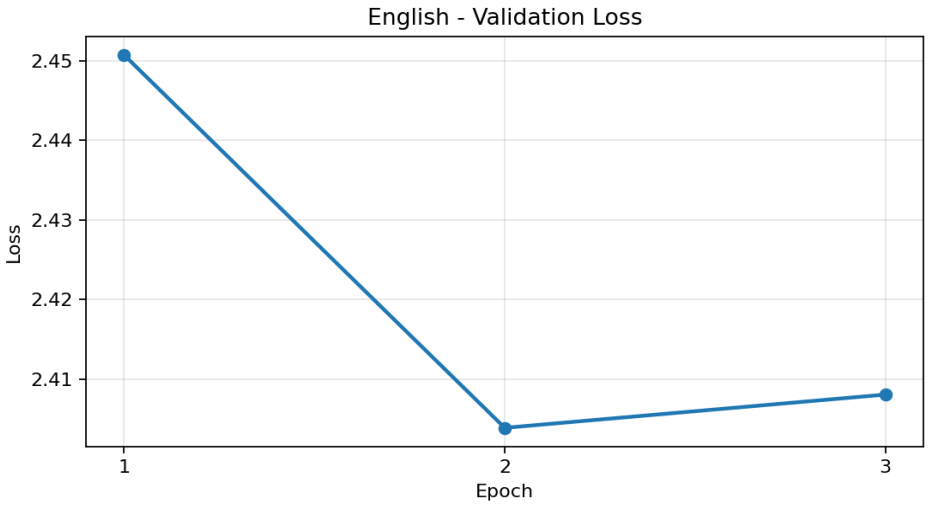
Training used PyTorch on CPU, Adam optimizer, learning rate 0.001, batch size 1, three epochs per language and a 90/10 split. The training objective combined mel reconstruction loss with duration, pitch and energy prediction losses.

6. Training Results

English

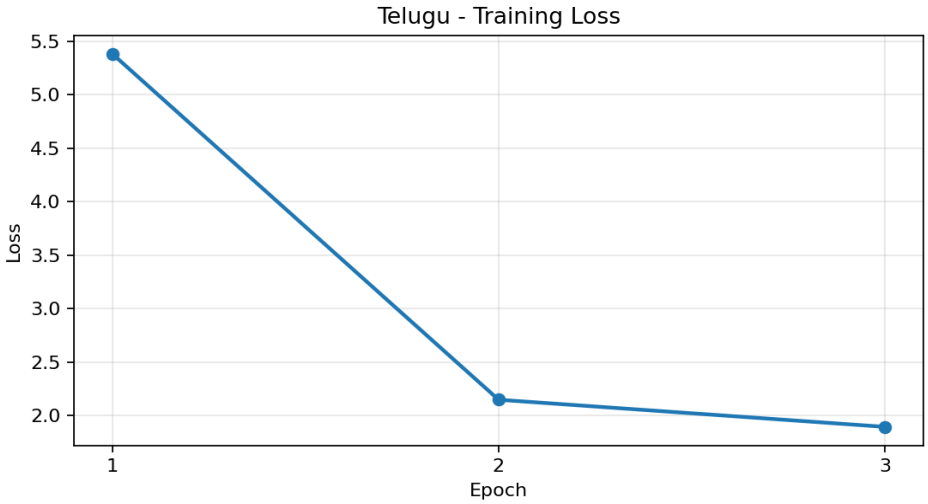
Epoch	Train Loss	Validation Loss
1	4.8397	2.4508
2	2.3102	2.4039
3	2.3021	2.4081

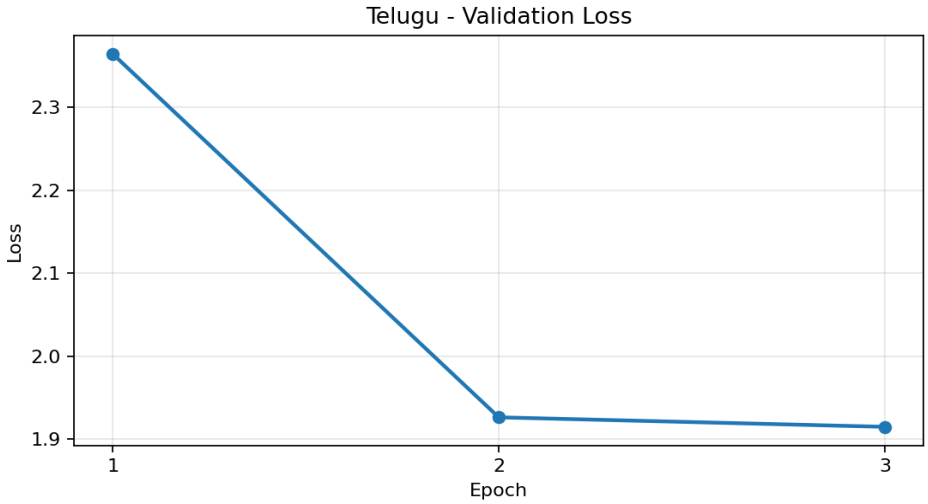




Telugu

Epoch	Train Loss	Validation Loss
1	5.3823	2.3646
2	2.1494	1.9263
3	1.8961	1.9148





7. Inference and Generated Audio

Five test utterances were synthesized for each language. Predicted mel-spectrograms were converted to WAV audio using Griffin-Lim reconstruction. Duration, pitch and energy prediction plots were saved for all ten test utterances. Audio generation was completed, but intelligibility and naturalness have not been formally evaluated.

8. Limitations

This implementation uses 100 samples per language, approximate phoneme alignment, a simplified model, and Griffin-Lim instead of a trained neural vocoder. It does not satisfy a multi-hour training-data requirement. No MOS, WER or CER evaluation was conducted. The generated speech may contain artifacts or unintelligible segments.

9. Conclusion

The project successfully builds an end-to-end bilingual FastSpeech2-style prototype with preprocessing, That covers model training, checkpointing, mel prediction, audio reconstruction and diagnostic plots. Further work should use larger aligned datasets, more complete FastSpeech2 training and a neural vocoder.